

Before starting this exercise sheet, please create the following two packages:

- 1) entities
- 2) apps

A) Building basic classes

Exercise A1) Person:

Create a class called **Person** in a file called **Person.java**. This should be created in the **entities** package.

A Person should contain 4 attributes (variables):

1. First name
2. Second name
3. Age
4. Whether or not they're left-handed (true/false)

You can set the default values of these to be anything you want.

Commit your code after your class is complete.

Exercise A2) Accessing an object's attributes:

Write a program called **PersonTestBed** in a file called **PersonTestBed.java**. This should be created in the **apps** package. Your program should:

1. Import your Person class.
2. Create a Person object
 - a. Note: you will need to use **Person variable_name = new Person()** to do this
3. Display the Person's full name in a formatted manner based on their left-handedness status
 - a. If they are left-handed, the text should be displayed normally.
 - b. If they are not left-handed, the text should be displayed in all caps.

Exercise A3) Setting/changing an object's attributes:

Amend your program from A2 to create a second Person object (p2). Set the data inside to user-supplied values (i.e. ask the user to enter each of the values and store them within p2's attributes/variables by editing them directly)

Once the user has supplied all the values required, display their full name in the same formatted manner as used for your Person from A2, i.e. display their name in all caps if they are left-handed, and display it normally otherwise.

Commit your code after your program is complete.

B) Adding methods

*When attempting this section, remember that currently we **do not want static methods** in our classes – we want every instance (object) we make to have its own version of the method!*

Exercise B1) Rectangle:

Create a class called **Rectangle** in **Rectangle.java**. This should be created in the **entities** package.

A Rectangle should contain 3 attributes (instance variables):

1. Length
2. Width
3. Colour

You can set the default values of these to be anything you want.

Exercise B2) Creating a display method:

Create a method within your Rectangle class to display its information. This method should be called **display()** and should print the values of the attributes in the format:

"Rectangle{length=X, width=Y, colour=Z}" where X, Y and Z are the values of the three attributes.

Commit your code after your class is complete, including your method.

Exercise B3) Calculating the area of a rectangle:

Create a method within your Rectangle class called **calcArea()** that will calculate its area (length * width) & return it.

Exercise B4) Using your class:

Remember to import your Rectangle class!

Write a program called **RectangleTestBed** in a file called **RectangleTestBed.java**. This file should be created in the **apps** package. This program should ask a user to create 5 Rectangle objects & store them in an arraylist. For each, your program should:

1. Create the current Rectangle
2. Ask the user for this Rectangle's length, width and colour.
3. Set these values in the current rectangle.
4. Add this Rectangle to the list.

Once your list contains 5 Rectangles, your program should:

- Locate the Rectangle with the largest area and display its details using its display method.
- Locate the Rectangle with the smallest width and display its position in the list.
- Locate all Rectangles with the colour "red" (case-insensitive), store them in a new list & display them once all Rectangles have been processed. If there are no red Rectangles, display a message indicating this to the user.
- Ask the user to enter a colour, then repeat the above to find, store & display all Rectangles of that colour.

Note: You should consider how you might use functions to improve your code.

Commit your code:

- *When your method has been added*
- *When the user can create rectangles*
- *When your program can search for rectangles based on different characteristics*